7 I/O Commands

An I/O command is a command submitted to an I/O Submission Queue. Figure 589 lists the I/O commands that are defined for use in all I/O Command Sets. The following subsections provide definitions for each command. Refer to section 3.1.3.4 for mandatory, optional, and prohibited I/O commands for the various controller types. The following subsections describe the definition for each of these commands.

The user data format and any end-to-end protection information is I/O Command Set specific. Refer to each I/O Command Set specification for applicability and additional details, if any. Refer to the referenced NVM Express I/O Command Set specification for all I/O Command Set specific commands described in Figure 589.

Commands shall only be submitted by the host when the controller is ready as indicated in the Controller Status property (CSTS.RDY) and after appropriate I/O Submission Queue(s) and I/O Completion Queue(s) have been created.

The submission queue entry (SQE) structure and the fields that are common to all I/O commands are defined in section 4.1. The completion queue entry (CQE) structure and the fields that are common to all I/O commands are defined in section 4.2. The command specific fields in the SQE and CQE structures (i.e., SQE Command Dwords 10-15, CQE Dword 0, and CQE Dword 1) for I/O commands supported across all I/O Command Sets are defined in this section.

Figure 589: Opcodes for I/O Commands

Opcode by Field		Combined Opcode[1]	Command[2]	Reference
(07:02)	(01:00)			
Function	Data Transfer[3]			
0000 00b	00b	00h	Flush[4]	7.2
0000 11b	01b	0Dh	Reservation Register	7.6
0000 11b	10b	0Eh	Reservation Report	7.8
0001 00b	01b	11h	Reservation Acquire	7.5
0001 00b	10b	12h	I/O Management Receive	7.3
0001 01b	01b	15h	Reservation Release	7.7
0001 10b	00b	18h	Cancel[4]	7.1
0001 11b	01b	1Dh	I/O Management Send	7.4
0111 11b	11b[5]	7Fh	Fabric Commands[5]	6
Vendor Specific				
1xxx xxb	NOTE 3	80h to FFh	Vendor specific	

Notes:
1. Opcodes not listed are I/O Command Set specific or reserved. Refer to Figure 91 for Opcode details.
2. All I/O commands use the Namespace Identifier (NSID) field. The value FFFFFFFFh is not supported in this field unless footnote 4 in this figure indicates that a specific command does support that value.
3. 00b = no data transfer; 01b = host to controller; 10b = controller to host; 11b = bidirectional. Refer to the Transfer Direction field in Figure 91.
4. This command may support the use of the Namespace Identifier (NSID) field set to FFFFFFFFh.
5. All Fabrics commands use the opcode 7Fh with the direction of data transfer specified as shown in Figure 574. Refer to section 6 for details.

7.1 Cancel command

The Cancel command is used to request an abort for specified commands submitted on the same I/O Submission Queue to which the Cancel command is submitted. The Cancel command may apply to a single namespace or to multiple namespaces. The Cancel command may be deeply queued. Some of the commands to abort may have already completed, currently be processing, or be deeply queued. As a result, the performance of the Cancel command may be impacted.

To abort an Admin command, the host uses the Abort command (refer to section 5.2.1). To abort a Cancel command, the host may:

- use the Abort command;
- use a second Cancel command with a Cancel Action set to Single Command Cancel (i.e., 00b) that specifies the CID of the Cancel command to abort; or
- follow the procedure described in section 3.3.1.3 to delete the I/O Submission Queue and recreate the I/O Submission Queue.

The controller applies the specified Cancel Action (refer to Figure 591) to all outstanding commands that have been fetched prior to the processing of the Cancel command. The controller may or may not apply the specified Cancel Action to commands that are fetched:

- after the start of processing of the Cancel command; and
- before the completion of the Cancel command.

There is no requirement to apply the specified Cancel Action to commands that have not been fetched by the controller.

If an abort action is performed on a command to abort, that action may be:

- an immediate abort (i.e., the abort action occurs prior to posting the completion queue entry for the Cancel command); or
- a deferred abort (i.e., the abort action may or may not occur prior to posting the completion queue entry for the Cancel command).

For each command on which an immediate abort is performed, the controller shall meet the immediate abort requirements (refer to section 5.2.1.1) for that command before posting the completion queue entry for the Cancel command.

For each command on which a deferred abort is performed, there are no requirements on the ordering of posting of the completion queue entry for the Cancel command (refer to section 7.1.1).

For each command on which an abort action is performed (i.e., immediate abort or deferred abort) the status code shall be set to Command Abort Requested in the completion queue entry for that command.

If the Cancel command is supported, then the Commands Supported and Effects log page shall be supported.

The Cancel command uses the Command Dword 10 and Command Dword 11 fields. All other command specific fields are reserved.

Figure 590: Cancel – Command Dword 10

Bits	Description
31:16	**Command Identifier (CID):** This field specifies the command identifier of the command to be aborted (i.e., the CDW0.CID field within the command to be aborted). If the Action Code field is set to Single Command Cancel and this field is set to the CID for this Cancel command, then the controller shall abort the command with a status code of Invalid Command ID. If the Action Code field is set to Multiple Command Cancel and this field is not set to FFFFh, then the controller shall abort the command with a status code of Invalid Field in Command.
15:00	**Submission Queue Identifier (SQID):** This field specifies the identifier of the Submission Queue that the Cancel command is associated with. If the SQID does not match the Submission Queue Identifier of the submission queue to which the Cancel command is submitted, then the controller shall abort the command with a status code of Invalid Field in Command.

Figure 591: Cancel – Command Dword 11

Bits	Description	
31:02	Reserved	
01:00	**Action Code (ACODE):**	
	Value	**Cancel Action**
	00b	**Single Command Cancel:** The hosts requests that the controller abort the specific command submitted to the specified namespace with the specified CID. If the NSID field is set to FFFFFFFFh then, the host requests that the controller abort the specific command submitted to any NSID with the specified CID. If the NSID is not set to FFFFFFFFh and the specified CID is not associated with the specified NSID then, the controller shall abort the Cancel command with a status code of Invalid Field in Command. If the specified CID is not found, then the controller shall complete the Cancel command with the Commands Eligible for Deferred Abort field cleared to 0h and the Commands Aborted field cleared to 0h.
	01b	**Multiple Command Cancel:** The hosts requests that the controller abort all commands, other than this Cancel command, on the I/O Submission Queue to which the Cancel command was submitted that were submitted to the specified NSID. If the NSID is set to FFFFFFFFh then the host requests that the controller abort all commands, other than this Cancel command, submitted to the I/O Submission Queue to which the Cancel command was submitted for all namespaces.
	All others	Reserved

7.1.1 Command Completion

Upon completion of the Cancel command, the controller posts a completion queue entry to the I/O Completion Queue indicating the status for the Cancel command.

If the Cancel Action (refer to Figure 591) specified a Single Command Cancel Action and the Commands Aborted field is cleared to 0h, then the host should examine the status in the completion queue entry of the command to abort to determine whether the command was aborted or not (i.e., whether a deferred abort was performed or not).

If the Cancel Action specified a Multiple Command Cancel Action, then the host should examine the status in the completion queue entry of each command to abort to determine whether the command was aborted or not.

Cancel command specific status code values are defined in Figure 592.

Figure 592: Cancel – Command Specific Status Values

Value	Definition
84h	**Invalid Command ID:** The specified CID matched the CID of this Cancel command.

Dword 0 of the completion queue entry contains information about the number of commands that were aborted by this command. Dword 0 of the completion queue entry is defined in Figure 593.

Figure 593: Cancel – Completion Queue Entry Dword 0

Bits	Description
31:16	**Commands Eligible for Deferred Abort (CEDA):** This field indicates the number of commands that match the specified criteria (refer to Figure 590 and Figure 591) on which a deferred abort may be performed. A value of 0h indicates that no commands are eligible for a deferred abort or that the controller does not support deferred aborts. A value of FFFFh indicates that FFFFh or more commands are eligible for a deferred abort. If the Cancel Action (refer to Figure 591) specified a Single Command Cancel and an immediate abort was performed on the specified command, then this field shall be cleared to 0h.
15:00	**Commands Aborted (CMDA):** This field indicates the number of commands on which the controller performed an immediate abort as a result of processing this Cancel command. A value of 0h indicates that the controller did not perform an immediate abort on any commands as a result of processing this Cancel command. If the Cancel Action (refer to Figure 591) specified a Single Command Cancel and an immediate abort was performed on the specified command, then this field shall be set to 1h.

7.2 Flush command

The Flush command is used to request that the contents of volatile write cache be made non-volatile.

If a volatile write cache is enabled (refer to section 5.2.26.1.4), then the Flush command shall commit data and metadata associated with the specified namespace(s) to non-volatile storage media. The flush applies to all commands for the specified namespace(s) completed by the controller prior to the submission of the Flush command. The controller may also flush additional data and/or metadata from any namespace.

If the Flush Behavior (FB) field is set to 11b in the VWC field in the Identify Controller data structure (refer to Figure 328) and the specified NSID is FFFFFFFFh, then the Flush command applies to all namespaces attached to the controller processing the Flush command. If the FB field is set to 10b and the specified NSID is FFFFFFFFh, then the controller aborts the command with a status code of Invalid Namespace or Format. If the FB field is cleared to 00b, then the controller behavior if the specified NSID is FFFFFFFFh is not indicated. Controllers compliant with NVM Express Base Specification, Revision 1.4 and later shall not set the FB field to the value of 00b.

If a namespace exists in an Endurance Group that has:

- Flexible Data Placement (refer to section 8.1.11) enabled; and
- the Volatile Write Cache Not Present (VWCNP) bit set to ‘1’ in the I/O Command Set Independent Identify Namespace data structure (refer to Figure 335),

then, even though the Volatile Write Cache field in the Identify Controller data structure (refer to Figure 328) indicates the presence of a volatile write cache in the controller, this namespace does not have a volatile write cache present.

A host may use the Volatile Write Cache Not Present (VWCNP) bit in the I/O Command Set Independent Identify Namespace data structure to determine if a Volatile Write Cache is not present for a namespace.

If a volatile write cache is not present or not enabled, then Flush commands shall have no effect and:

a) shall complete successfully if a sanitize operation is not in progress; and
b) may complete successfully if a sanitize operation is in progress.

All command specific fields are reserved.

7.2.1 Command Completion

Upon completion of the Flush command, the controller posts a completion queue entry to the associated I/O Completion Queue.

7.3 I/O Management Receive command

The I/O Management Receive command is used to receive information from the controller used by the host to manage I/O. The behavior of the command is dependent on the specified operation as defined in the Management Operation field in Figure 595.

The command uses the Data Pointer, Command Dword 10, and Command Dword 11 fields. All other command specific fields are reserved. If the command uses PRPs for the data transfer, then the PRP Entry 1 and PRP Entry 2 fields are used. If the command uses SGLs for the data transfer, then the SGL Entry 1 field is used. All other command specific fields are reserved.

If the Number of Dwords (NUMD) field corresponds to a length that is less than the size of the data structure to be returned, then only that specified portion of the data structure is transferred. If the NUMD field corresponds to a length that is greater than the size of the associated data structure, then the entire contents of the data structure are transferred and no additional data is transferred, unless otherwise specified.

Figure 594: I/O Management Receive – Data Pointer

Bits	Description
127:00	**Data Pointer (DPTR):** This field specifies the location of a data buffer where data is transferred from. Refer to Figure 92 for the definition of this field.

Figure 595: I/O Management Receive – Command Dword 10

<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>31:16</td><td>Management Operation Specific (MOS): This definition of this field is specific the value of the MO field. If this field is not defined for the management operation specified by the MO field, then this field is reserved.</td></tr>
<tr><td>15:08</td><td>Reserved</td></tr>
<tr><td>07:00</td><td>Management Operation (MO): This field specifies the management operation to perform.
<table>
<tr><th>Value</th><th>Definition</th></tr>
<tr><td>00h</td><td>No action</td></tr>
<tr><td>01h</td><td>Reclaim Unit Handle Status: For each Placement Handle of the namespace, the controller shall return a Reclaim Unit Handle Status Descriptor for each Reclaim Group.</td></tr>
<tr><td>FFh</td><td>Vendor specific</td></tr>
<tr><td>All others</td><td>Reserved</td></tr>
</table></td></tr>
</table>

Figure 596: I/O Management Receive – Command Dword 11

Bits	Description
31:00	**Number of Dwords (NUMD):** This field specifies the number of dwords to transfer. This is a 0's based value.

7.3.1 I/O Management Receive Operations

7.3.1.1 Reclaim Unit Handle Status (Management Operation 01h)

The Reclaim Unit Handle Status management operation is used to provide information about Reclaim Unit Handles (refer to section 1.5.89) that are accessible by the specified namespace. The returned data contains one or more Reclaim Unit Handle Status Descriptor data structures (refer to the applicable NVM Express I/O Command Set specification). The information contained in each Reclaim Unit Handle Status Descriptor:

- is the information at the time the controller processes that Reclaim Unit Handle Status Descriptor; and
- may or may not contain the information reflecting any outstanding command that affects the reported Reclaim Unit Handle Status Descriptor.

If Flexible Data Placement is disabled in the Endurance Group containing the specified namespace, then the command shall be aborted with a status code of FDP Disabled.

If the NSID field is set to 0h or FFFFFFFFh, then the controller shall abort the command with a status code of Invalid Namespace or Format.

If the Number of Dwords (NUMD) field corresponds to a length that is greater than the size of the Reclaim Unit Handle Status data structure (refer to Figure 597), then the entire contents of the data structure are transferred and zeroes are transferred beyond the end of that data structure.

Figure 597: Reclaim Unit Handle Status

Bytes	Description
Header	
13:00	Reserved
15:14	**Number of Reclaim Unit Handle Status Descriptors (NRUHSD):** This field indicates the number of Reclaim Unit Handle Status Descriptors in the Reclaim Unit Handle Status Descriptor list.
Reclaim Unit Handle Status Descriptor List	
47:16	**Reclaim Unit Handle Status Descriptor 1:** The first Reclaim Unit Handle Status Descriptor (refer to the applicable NVM Express I/O Command Set specification).
79:48	**Reclaim Unit Handle Status Descriptor 2:** The second Reclaim Unit Handle Status Descriptor (refer to the applicable NVM Express I/O Command Set specification), if any.
…	…
(NRUHSD *32)+15: (NRUHSD *32)–16	**Reclaim Unit Handle Status Descriptor NRUHSD:** The last Reclaim Unit Handle Status Descriptor (refer to the applicable NVM Express I/O Command Set specification), if any.

The Reclaim Unit Handle Status Descriptors are defined in the NVM Express I/O Command Set specifications, if supported.

7.3.2 Command Completion

When the command is completed with success or failure, the controller shall post a completion queue entry to the associated I/O Completion Queue indicating the status for the command.

7.4 I/O Management Send command

The I/O Management Send command is used to manage I/O and the behavior of the command is dependent on the specified operation as defined in the Management Operation field in Figure 599.

The command uses the Data Pointer and Command Dword 10 field. All other command specific fields are reserved. If the command uses PRPs for the data transfer, then the PRP Entry 1 and PRP Entry 2 fields are used. If the command uses SGLs for the data transfer, then the SGL Entry 1 field is used. All other command specific fields are reserved.

Figure 598: I/O Management Send – Data Pointer

Bits	Description
127:00	**Data Pointer (DPTR):** This field specifies the location of a data buffer where data is transferred from. Refer to Figure 92 for the definition of this field.

Figure 599: I/O Management Send – Command Dword 10

Bits	Description
31:16	**Management Operation Specific (MOS):** The definition of this field is specific to the value of the MO field. If this field is not defined for the management operation specified by the MO field, then this field is reserved.
15:08	Reserved

Figure 599: I/O Management Send – Command Dword 10

<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>07:00</td><td>Management Operation (MO): This field specifies the management operation to perform.
<table>
<tr><th>Value</th><th>Definition</th></tr>
<tr><td>00h</td><td>No action</td></tr>
<tr><td>01h</td><td>Reclaim Unit Handle Update: Update the reference to the Reclaim Unit specified by each entry in the Placement Identifier list to reference an empty Reclaim Unit (refer to section 3.2.4).</td></tr>
<tr><td>FFh</td><td>Vendor specific</td></tr>
<tr><td>All others</td><td>Reserved</td></tr>
</table>
</td></tr>
</table>

7.4.1 I/O Management Send Operations

7.4.1.1 Reclaim Unit Handle Update (Management Operation 01h)

The Reclaim Unit Handle Update management operation for the I/O Management Send command provides a list of Placement Identifiers (refer to Figure 601). The number of Placement Identifiers is defined in the Management Operation Specific field defined in Figure 600. For each Placement Identifier in the list:

- If the currently referenced Reclaim Unit has been written with user data, then the Placement Identifier shall be modified to reference a different Reclaim Unit that is empty (refer to section 3.2.4); and
- If the currently referenced Reclaim Unit has not been written with any user data (i.e., is already empty), then the Placement Identifier may be modified to reference a different Reclaim Unit that is empty.

Figure 600: Management Operation Specific – Reclaim Unit Handle Update Operation

<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:00</td><td>Number of Placement Identifiers (NPID): Indicates the number of Placement Identifiers that are specified in the command. This is a 0’s based value.

This field shall not exceed the minimum of:
the value in the Max Placement Identifiers (MAXPIDS) field of the enabled FDP configuration (refer to Figure 287); and
the product of the Number of Reclaim Groups (NRG) field and the Number of Reclaim Unit Handles (NRUH) field of the enabled FDP configuration (refer to Figure 287);</td></tr>
</table>

If a specified Placement Identifier is invalid due to:

- the value of the Reclaim Group Identifier field being greater than or equal to the Number of Reclaim Groups field of the FDP Configuration Descriptor (refer to Figure 287) for the Endurance Group associated with the specified namespace; or
- the specified Placement Handle field being greater than or equal to the Number of Placement Handles field specified when the namespace was created,

then the controller shall abort the command with a status code of Invalid Field in Command.

If the value represented by the Number of Placement Identifiers (NPID) field is greater than the Max Placement Identifiers (MAXPIDS) field (refer to Figure 287) in the current FDP configuration, then the controller shall abort the command with a status code of Invalid Field in Command.

If the command is aborted, then Placement Identifiers may or may not have been updated.

While processing an I/O Management Send command that specifies the Reclaim Unit Handle Update operation, if the controller processes a write command that utilizes a Placement Identifier specified in the Placement Identifier List of that I/O Management Send command, then the controller may write the user data for that write command to the referenced Reclaim Unit:

- prior to processing that I/O Management Send command; or
- upon the completion of that I/O Management Send command.

Figure 601: Reclaim Unit Handle Update – Data Buffer

Bytes	Description
Placement Identifier List	
01:00	**Placement Identifier 1:** This field specifies the first Placement Identifier that indicates a Placement Handle and a Reclaim Group Identifier. Refer to Figure 289 and Figure 290.
03:02	**Placement Identifier 2:** This field specifies the second Placement Identifier that indicates a Placement Handle and a Reclaim Group Identifier, if any. Refer to Figure 289 and Figure 290.
…	
(NPID*2)+1:(NPID*2)	**Placement Identifier NPID:** This field specifies the last Placement Identifier that indicates a Placement Handle and a Reclaim Group Identifier, if any. Refer to Figure 289 and Figure 290.

If Flexible Data Placement is disabled in the Endurance Group containing the specified namespace, then the controller shall abort the command with a status code of FDP Disabled.

7.4.2 Command Completion

When the command is completed with success or failure, the controller shall post a completion queue entry to the associated I/O Completion Queue indicating the status for the command.

7.5 Reservation Acquire command

The Reservation Acquire command is used to:

- acquire a reservation on a namespace;
- preempt a reservation held on a namespace; or
- preempt a reservation held on a namespace and abort outstanding commands for that namespace.

The command uses Command Dword 10 and a Reservation Acquire data structure in memory. If the command uses PRPs for the data transfer, then PRP Entry 1 and PRP Entry 2 fields are used. If the command uses SGLs for the data transfer, then the SGL Entry 1 field is used. All other command specific fields are reserved.

Figure 602: Reservation Acquire – Data Pointer

Bits	Description
127:00	**Data Pointer (DPTR):** This field specifies the location of a data buffer where data is transferred from. Refer to Figure 92 for the definition of this field.

Figure 603: Reservation Acquire – Command Dword 10

Bits	Description
31:16	Reserved
15:08	**Reservation Type (RTYPE):** This field specifies the type of reservation to be created. The field is defined in Figure 605.
07:05	Reserved

Figure 603: Reservation Acquire – Command Dword 10

<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>04</td><td>Dispersed Namespace Reservation Support (DISNSRS): This bit specifies host support for reservations on dispersed namespaces for a host that supports dispersed namespaces (i.e., for a host that has set the Host Dispersed Namespace Support (HDISNS) field to 1h in the Host Behavior Support feature).

If this bit is set to ‘1’, then the host supports reservations on dispersed namespaces (i.e., the host supports receiving a value of FFFDh in the Controller ID (CNTLID) field of a Registered Controller data structure or Registered Controller Extended data structure (refer to section 8.1.10.6)).

If this bit is cleared to ‘0’, then the host does not support reservations on dispersed namespaces (i.e., the host does not support receiving a value of FFFDh in the Controller ID (CNTLID) field of a Registered Controller data structure or Registered Controller Extended data structure (refer to section 8.1.10.6)). If the HDISNS field is set to 1h in the Host Behavior Support feature and the host submits the command to a dispersed namespace with this bit cleared to ‘0’, then the controller aborts the command with a status code of Namespace Is Dispersed as described in section 8.1.10.6.</td></tr>
<tr><td>03</td><td>Ignore Existing Key (IEKEY): If this bit is set to a ‘1’, then the controller shall return an error of Invalid Field in Command. If this bit is cleared to ‘0’, then the Current Reservation Key is checked.</td></tr>
<tr><td>02:00</td><td>Reservation Acquire Action (RACQA): This field specifies the action that is performed by the command.
<table><tr><th>Value</th><th>Definition</th><th>Reference</th></tr><tr><td>000b</td><td>Acquire</td><td>8.1.24.5</td></tr><tr><td>001b</td><td>Preempt</td><td>8.1.24.7</td></tr><tr><td>010b</td><td>Preempt and Abort</td><td>8.1.24.7</td></tr><tr><td>011b to 111b</td><td colspan="2">Reserved</td></tr></table></td></tr>
</table>

Figure 604: Reservation Acquire Data Structure

Bytes	Description
07:00	**Current Reservation Key (CRKEY):** The field specifies the current reservation key associated with the host.
15:08	**Preempt Reservation Key (PRKEY):** If the Reservation Acquire Action is set to 001b (i.e., Preempt) or 010b (i.e., Preempt and Abort), then this field specifies the reservation key to be unregistered from the namespace. For all other Reservation Acquire Action values, this field is reserved.

Figure 605: Reservation Type Encoding

Value	Definition
0h	Reserved
1h	Write Exclusive Reservation
2h	Exclusive Access Reservation
3h	Write Exclusive - Registrants Only Reservation
4h	Exclusive Access - Registrants Only Reservation
5h	Write Exclusive - All Registrants Reservation
6h	Exclusive Access - All Registrants Reservation
7h to FFh	Reserved

7.5.1 Command Completion

When the command is completed, the controller shall post a completion queue entry to the associated I/O Completion Queue indicating the status for the command.

7.6 Reservation Register command

The Reservation Register command is used to register, unregister, or replace a reservation key.

The command uses Command Dword 10 and a Reservation Register data structure in memory (refer to Figure 608). If the command uses PRPs for the data transfer, then PRP Entry 1 and PRP Entry 2 fields are used. If the command uses SGLs for the data transfer, then the SGL Entry 1 field is used. All other command specific fields are reserved.

Figure 606: Reservation Register – Data Pointer

Bits	Description
127:00	**Data Pointer (DPTR):** This field specifies the location of a data buffer where data is transferred from. Refer to Figure 92 for the definition of this field.

Figure 607: Reservation Register – Command Dword 10

<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>31:30</td><td>Change Persist Through Power Loss State (CPTPL): This field allows the Persist Through Power Loss (PTPL) state associated with the namespace to be modified as a side effect of processing this command. If the Reservation Persistence feature (refer to section 5.2.26.1.34) is saveable, then any change to the PTPL state as a result of processing this command shall be applied to both the current value and the saved value of that feature.
<table>
<tr><th>Value</th><th>Definition</th></tr>
<tr><td>00b</td><td>No change to PTPL state</td></tr>
<tr><td>01b</td><td>Reserved</td></tr>
<tr><td>10b</td><td>Clear PTPL state to ‘0’. Reservations are released and registrants are cleared on a power on.</td></tr>
<tr><td>11b</td><td>Set PTPL state to ‘1’. Reservations and registrants persist across a power loss.</td></tr>
</table></td></tr>
<tr><td>29:05</td><td>Reserved</td></tr>
<tr><td>04</td><td>Dispersed Namespace Reservation Support (DISNSRS): This bit specifies host support for reservations on dispersed namespaces for a host that supports dispersed namespaces (i.e., for a host that has set the Host Dispersed Namespace Support (HDISNS) field to 1h in the Host Behavior Support feature).

If this bit is set to ‘1’, then the host supports reservations on dispersed namespaces (i.e., the host supports receiving a value of FFFDh in the Controller ID (CNTLID) field of a Registered Controller data structure or Registered Controller Extended data structure (refer to section 8.1.10.6)).

If this bit is cleared to ‘0’, then the host does not support reservations on dispersed namespaces (i.e., the host does not support receiving a value of FFFDh in the Controller ID (CNTLID) field of a Registered Controller data structure or Registered Controller Extended data structure (refer to section 8.1.10.6)). If the HDISNS field is set to 1h in the Host Behavior Support feature and the host submits the command to a dispersed namespace with this bit cleared to ‘0’, then the controller aborts the command with a status code of Namespace Is Dispersed as described in section 8.1.10.6.</td></tr>
<tr><td>03</td><td>Ignore Existing Key (IEKEY): If this bit is set to a ‘1’, then Reservation Register Action (RREGA) field values that use the Current Reservation Key (CRKEY) shall succeed regardless of the value of the Current Reservation Key field in the command (i.e., the current reservation key, if any, is not checked, and absence of a current reservation key does not cause an error).</td></tr>
<tr><td>02:00</td><td>Reservation Register Action (RREGA): This field specifies the registration action that is performed by the command.
<table>
<tr><th>Value</th><th>Definition</th><th>Reference</th></tr>
<tr><td>000b</td><td>Register Reservation Key</td><td>8.1.24.3</td></tr>
<tr><td>001b</td><td>Unregister Reservation Key</td><td>8.1.24.4</td></tr>
<tr><td>010b</td><td>Replace Reservation Key</td><td>8.1.24.3</td></tr>
<tr><td>011b to 111b</td><td colspan="2">Reserved</td></tr>
</table></td></tr>
</table>

Figure 608: Reservation Register Data Structure

Bytes	Description
07:00	**Current Reservation Key (CRKEY):** If the Reservation Register Action is 001b (i.e., Unregister Reservation Key) or 010b (i.e., Replace Reservation Key), then this field contains the current reservation key associated with the host. For all other Reservation Register Action values, this field is reserved. The controller ignores the value of this field when the Ignore Existing Key (IEKEY) bit is set to ‘1’.

Figure 608: Reservation Register Data Structure

Bytes	Description
15:08	**New Reservation Key (NRKEY):** If the Reservation Register Action field is cleared to 000b (i.e., Register Reservation Key) or 010b (i.e., Replace Reservation Key), then this field contains the new reservation key associated with the host. For all other Reservation Register Action values, this field is reserved.

7.6.1 Command Completion

When the command is completed, the controller shall post a completion queue entry to the associated I/O Completion Queue indicating the status for the command.

7.7 Reservation Release command

The Reservation Release command is used to release or clear a reservation held on a namespace.

The command uses Command Dword 10 and a Reservation Release data structure in memory. If the command uses PRPs for the data transfer, then PRP Entry 1 and PRP Entry 2 fields are used. If the command uses SGLs for the data transfer, then the SGL Entry 1 field is used. All other command specific fields are reserved.

Figure 609: Reservation Release – Data Pointer

Bits	Description
127:00	**Data Pointer (DPTR):** This field specifies the location of a data buffer where data is transferred from. Refer to Figure 92 for the definition of this field.

Figure 610: Reservation Release – Command Dword 10

Bits	Description
31:16	Reserved
15:08	**Reservation Type (RTYPE):** If the Reservation Release Action field is cleared to 000b (i.e., Release), then this field specifies the type of reservation that is being released. The reservation type in this field shall match the current reservation type. If the reservation type in this field does not match the current reservation type, then the controller should return a status code of Invalid Field in Command. This field is defined in Figure 605.
07:05	Reserved
04	**Dispersed Namespace Reservation Support (DISNSRS):** This bit specifies host support for reservations on dispersed namespaces for a host that supports dispersed namespaces (i.e., for a host that has set the Host Dispersed Namespace Support (HDISNS) field to 1h in the Host Behavior Support feature). If this bit is set to ‘1’, then the host supports reservations on dispersed namespaces (i.e., the host supports receiving a value of FFFDh in the Controller ID (CNTLID) field of a Registered Controller data structure or Registered Controller Extended data structure (refer to section 8.1.10.6)). If this bit is cleared to ‘0’, then the host does not support reservations on dispersed namespaces (i.e., the host does not support receiving a value of FFFDh in the Controller ID (CNTLID) field of a Registered Controller data structure or Registered Controller Extended data structure (refer to section 8.1.10.6)). If the HDISNS field is set to 1h in the Host Behavior Support feature and the host submits the command to a dispersed namespace with this bit cleared to ‘0’, then the controller aborts the command with a status code of Namespace Is Dispersed as described in section 8.1.10.6.
03	**Ignore Existing Key (IEKEY)**: If this bit is set to a ‘1’, then the controller shall return an error of Invalid Field in Command. If this bit is cleared to ‘0’, then the Current Reservation Key is checked.
02:00	**Reservation Release Action (RRELA):** This field specifies the reservation action that is performed by the command. Value / Definition / Reference: 000b: Release (8.1.24.6) 001b: Clear (8.1.24.8) 010b to 111b: Reserved

Nested table in bits 02:00:

Value	Definition	Reference
000b	Release	8.1.24.6
001b	Clear	8.1.24.8
010b to 111b	Reserved	

Figure 611: Reservation Release Data Structure

Bytes	O/M	Description
7:0	M	**Current Reservation Key (CRKEY):** The field specifies the current reservation key associated with the host.

7.7.1 Command Completion

When the command is completed, the controller shall post a completion queue entry to the associated I/O Completion Queue indicating the status for the command.

7.8 Reservation Report command

The Reservation Report command returns a Reservation Status data structure to memory that describes the registration and reservation status of a namespace.

If the namespace is not a dispersed namespace, then the size of the Reservation Status data structure is a function of the number of registrants of the namespace (i.e., there is a Registrant data structure and/or Registrant Extended data structure for each such registrant). If the namespace is a dispersed namespace that is able to be accessed by controllers in multiple participating NVM subsystems, then the size of the Reservation Status data structure is a function of the number of registrants of the namespace in the NVM subsystem containing the controller processing the command and the number of registrants of the namespace in each separate participating NVM subsystem. The controller returns the data structure in Figure 615 if the host has selected a 64-bit Host Identifier and the data structure in Figure 616 if the host has selected a 128-bit Host Identifier (refer to section 5.2.26.1.32).

For controllers compliant with NVM Express Base Specification, Revision 2.0 and earlier, registrants of the namespace that are not associated with any controller in the NVM subsystem may or may not be reported by this command.

If a 64-bit Host Identifier has been specified and the Extended Data Structure bit is set to '1' in Command Dword 11, then the controller shall abort the command with the status code of Host Identifier Inconsistent Format. If a 128-bit Host Identifier has been specified and the Extended Data Structure bit is cleared to '0' in Command Dword 11, then the controller shall abort the command with the status code of Host Identifier Inconsistent Format.

The command uses Command Dword 10 and Command Dword 11. If the command uses PRPs for the data transfer, then PRP Entry 1 and PRP Entry 2 fields are used. If the command uses SGLs for the data transfer, then the SGL Entry 1 field is used. All other command specific fields are reserved.

Figure 612: Reservation Report – Data Pointer

Bits	Description
127:00	**Data Pointer (DPTR):** This field specifies the location of a data buffer where data is transferred to. Refer to Figure 92 for the definition of this field.

Figure 613: Reservation Report – Command Dword 10

Bits	Description
31:00	**Number of Dwords (NUMD):** This field specifies the number of dwords of the Reservation Status data structure to transfer. This is a 0's based value. If this field corresponds to a length that is less than the size of the Reservation Status data structure, then only that specified portion of the data structure is transferred. If this field corresponds to a length that is greater than the size of the Reservation Status data structure, then the entire contents of the data structure are transferred and no additional data is transferred.

Figure 614: Reservation Report – Command Dword 11

Bits	Description
31:02	Reserved
01	**Dispersed Namespace Reservation Support (DISNSRS):** This bit specifies host support for reservations on dispersed namespaces for a host that supports dispersed namespaces (i.e., for a host that has set the Host Dispersed Namespace Support (HDISNS) field to 1h in the Host Behavior Support feature). If this bit is set to '1', then the host supports reservations on dispersed namespaces (i.e., the host supports receiving a value of FFFDh in the Controller ID (CNTLID) field of a Registrant data structure or Registrant Extended data structure (refer to section 8.1.10.6)). If this bit is cleared to '0', then the host does not support reservations on dispersed namespaces (i.e., the host does not support receiving a value of FFFDh in the Controller ID (CNTLID) field of a Registrant data structure or Registrant Extended data structure (refer to section 8.1.10.6)). If the HDISNS field is set to 1h in the Host Behavior Support feature and the host submits the command to a dispersed namespace with this bit cleared to '0', then the controller aborts the command with a status code of Namespace Is Dispersed as described in section 8.1.10.6.
00	**Extended Data Structure (EDS):** If this bit is set to '1', then the controller returns the extended data structure defined in Figure 616. If this bit is cleared to '0', then the controller returns the data structure defined in Figure 615.

Figure 615: Reservation Status Data Structure

<table>
<tr><th>Bytes</th><th>Description</th></tr>
<tr><th colspan="2">Header</th></tr>
<tr><td>03:00</td><td>Generation (GEN): This field contains a 32-bit wrapping counter that is incremented any time any one the following occur:
a Reservation Register command completes successfully on any controller associated with the namespace;
a Reservation Release command with Reservation Release Action (RRELA) set to 001b (i.e., Clear) completes successfully on any controller associated with the namespace; and
a Reservation Acquire command with Reservation Acquire Action (RACQA) set to 001b (Preempt) or 010b (Preempt and Abort) completes successfully on any controller associated with the namespace.
If the value of this field is FFFFFFFFh, then the field shall be cleared to 0h when incremented (i.e., rolls over to 0h).</td></tr>
<tr><td>04</td><td>Reservation Type (RTYPE): This field indicates whether a reservation is held on the namespace. A value of 0h indicates that no reservation is held on the namespace. A non-zero value indicates a reservation is held on the namespace and the reservation type is defined in Figure 605.</td></tr>
<tr><td>06:05</td><td>Number of Registrants (REGSTRNT): This field indicates the number of registrants of the namespace. This indicates the number of Registrant data structures or Registrant Extended data structures contained in this data structure.

Note: This field was formerly named Number of Registered Controllers (REGCTL).</td></tr>
<tr><td>08:07</td><td>Reserved</td></tr>
<tr><td>09</td><td>Persist Through Power Loss State (PTPLS): This field indicates the Persist Through Power Loss State associated with the namespace.<table><tr><th>Value</th><th>Definition</th></tr><tr><td>0</td><td>Reservations are released and registrants are cleared on a power on.</td></tr><tr><td>1</td><td>Reservations and registrants persist across a power loss.</td></tr></table></td></tr>
<tr><td>23:10</td><td>Reserved</td></tr>
<tr><th colspan="2">Registered Controller Data Structure List</th></tr>
<tr><td>47:24</td><td>Registrant Data Structure 0

Note: This field was formerly named Registered Controller Data Structure 0.</td></tr>
<tr><td>…</td><td>…</td></tr>
</table>

Figure 615: Reservation Status Data Structure

Bytes	Description
24*n+47: 24*(n+1)	**Registrant Data Structure n**

Figure 616: Reservation Status Extended Data Structure

Bytes	Description
Header	
23:00	**Reservation Status Header (RSHDR):** Refer to the Reservation Status Header in Figure 615 for definition.
63:24	Reserved
Registered Controller Extended Data Structure List	
127:64	**Registrant Extended Data Structure 0** Note: This field was formerly named Registered Controller Extended Data Structure 0.
…	…
64*(n+1)+63: 64*(n+1)	**Registrant Extended Data Structure n**

Figure 617: Registered Controller Data Structure

<table>
<tr><th>Bytes</th><th>Description</th></tr>
<tr><td>01:00</td><td>Controller ID (CNTLID): If a registrant of the namespace is associated with a controller in the NVM subsystem, then this field contains the controller ID (i.e., the value of the CNTLID field in the Identify Controller data structure) of the controller that is associated with the registrant whose host identifier is indicated in the Host identifier field of this Registrant data structure.

If a registrant of the namespace is not associated with any controller in the NVM subsystem, then the controller processing the command shall set this field to FFFFh.

If the namespace is a dispersed namespace and the controller is not contained in the same participating NVM subsystem as the controller processing the command, then the Controller ID field is set to FFFDh, as described in section 8.1.10.6.</td></tr>
<tr><td>02</td><td>Reservation Status (RCSTS): This field indicates the reservation status of the registrant associated with this data structure (i.e., the registrant whose host identifier is indicated in the Host Identifier field of this Registrant data structure).
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>7:1</td><td>Reserved</td></tr>
<tr><td>0</td><td>Association with Host Holding Reservation (AHHR): If this bit is set to '1', then the registrant associated with this data structure holds a reservation on the namespace. If this bit is cleared to ‘0’, then the controller is not associated with a host that holds a reservation on the namespace.</td></tr>
</table></td></tr>
<tr><td>07:03</td><td>Reserved</td></tr>
<tr><td>15:08</td><td>Host Identifier (HOSTID): This field contains the 64-bit Host Identifier of the registrant of the namespace described by this data structure.</td></tr>
<tr><td>23:16</td><td>Reservation Key (RKEY): This field contains the reservation key of the registrant described by this data structure.</td></tr>
</table>

Figure 618: Registered Controller Extended Data Structure

Bytes	Description
01:00	**Controller ID (CNTLID):** Refer to Figure 617 for definition.
02	**Reservation Status (RCSTS):** Refer to Figure 617 for definition.
07:03	Reserved
15:08	**Reservation Key (RKEY):** Refer to Figure 617 for definition.

Figure 618: Registered Controller Extended Data Structure

Bytes	Description
31:16	**Host Identifier (HOSTID):** This field contains the 128-bit Host Identifier of the registrant of the namespace described by this data structure.
63:32	Reserved

7.8.1 Command Completion

When the command is completed, the controller shall post a completion queue entry to the associated I/O Completion Queue indicating the status for the command.